Paleomagnetic evidence for late Cenozoic glaciations in the Mackenzie Mountains of the Northwest Territories, Canada[1]

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Abstract: The Mackenzie Mountains were affected by montane valley glaciers during the Pleistocene and peripherally by the Laurentide Ice Sheet during the last glaciation. In this paper we report on magnetostratigraphic dating and correlation of three sections recording Late Pliocene to Late Pleistocene glaciations: Katherine Creek, Little Bear River, and Inlin Brook (located around 65°N, 127°W). Each section consists of a colluvial unit overlying a Pliocene pediment surface cut into Proterozoic or Paleozoic bedrock, or Tertiary gravel, which is in turn overlain by a stack of five, and in places six, montane tills, usually with soils developed at their surfaces, and capped by a Laurentide till. Normal and reversed magnetizations were recognized with single-domain magnetite as a dominant remanence carrier. The Katherine Creek section has a normally magnetized colluvium at its base, which is overlain by two reversed tills, succeeded by three normal tills. We interpret the top two tills to be of Brunhes age (<780 ka) but argue that the lowermost normal till is of probable Olduvai age (ca. 1.8 Ma). The two underlying tills are of Matuyama age (2.6 Ma to 780 ka), and the colluvial base is assigned to the Gauss (3.5–2.6 Ma). The Little Bear River section exposes a stratigraphic record similar to that found at Katherine Creek. Only four units could be assigned a paleomagnetic polarity, the others yielding incoherent results. Paleosols on the first and second till units were reversed and normal, respectively, and the top till was normal. Thus there is clear evidence of an older (reversed) Pleistocene glaciation and a magnetostratigraphic record compatible with that found at Katherine Creek. Magnetic measurements from Inlin Brook gave largely incoherent results, with the exception of the surface (Laurentide) till, which is normal. The glacial history recorded in the Mackenzie Mountains correlates well with other studies carried out in the Cordillera. The large-scale changes in climate revealed in these terrestrial records provide baseline data for paleoenvironmental reconstruction.

Résumé : Les monts Mackenzie subirent les effets des glaciers de vallées de montagne durant le Pléistocène, et leurs zones périphériques furent affectées par l'Inlandsis laurentidien durant la dernière glaciation. Ici, nous présentons les résultat d'études de datations magnétostratigraphiques et de la mise en corrélation de trois coupes qui documentent les glaciations apparues entre le Pliocène tardif et le Pléistocène tardif : Katherine Creek, Little Bear River et Inlin Brook (situées près de 65°N, 127°W). Chaque coupe est formée d'une unité colluviale qui recouvre une surface de pédiment pliocène taillée dans le substratum protérozoïque ou paléozoïque, ou de gravier tertiaire, et recouverte à son tour par un empilement de cinq, et à certains endroits de six, tills de montagne, qui exhibent habituellement à leurs surfaces des profils de sol développés, et enfin le tout est coiffé d'un till laurentidien. On a identifié des aimantations normales et inversées, le minéral porteur dominant de l'aimantation est une magnétite à domaine simple. La coupe de Katherine Creek est caractérisée par un dépôt de colluvion basal d'aimantation normale, lequel est recouvert par deux tills d'aimantation inversée, et ensuite de trois tills d'aimantation normale. Nous considérons que les deux tills au sommet datent de l'épisode de Brunhes (<780 ka), mais nous apportons des arguments permettent de croire que le till inférieur d'aimantation normale correspond à l'épisode d'Olduvai (environ 1,8 Ma). Les deux tills sous-jacents sont de l'épisode de Matuyama (2,6 Ma à 780 ka), et le dépôt de colluvion basal est assigné à l'épisode de Gauss (3,5 à 2,6 Ma). La coupe de Little Bear River expose une séquence stratigraphique analogue à celle exposée à Katherine Creek. Une polarité paléomagnétique ne peut être assignée qu'à seulement quatre des unités, pour les autres les résultats sont trop discordants. Les paléosols développés aux surfaces de la première et de la seconde unité de till présentent une aimantation inversée et normale respectivement, et le till sommital est caractérisé par une aimantation normale. Donc, il y a évidence d'une glaciation pléistocène plus ancienne (aimantation inversée), et existence d'un registre magnétostratigraphique compatible avec celui de Katherine Creek. Les mesures magnétiques d'Inlin Brook ont donné des résultats largement discordants, à l'exception du till de surface (Laurentidien) d'aimantation normale. L'histoire glaciaire enregistrée dans les monts Mackenzie est en excellente harmonie avec celle des autres régions étudiées dans la Cordillère. Les changements climatiques à grande échelle divulgués par ces registres terrestres fournissent les données de base pour le reconstitution du paléoenvironnement.
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Introduction

Late Cenozoic magnetostratigraphy has over the past 30 years been a very useful tool for the correlation and dating of terrestrial geologic deposits, and as such has served to establish time lines in sedimentary sequences, which contain proxy records of past climates (e.g., Birkeland et al. 1980; Easterbrook and Boellstorff 1982; Barendregt and Vincent 1990). Sedimentary deposits in the range of 1–3 Ma offer few other possibilities for absolute dating. Where suitable fine-grained sediments exist, their paleomagnetic polarity can be determined and, using ancillary information, correlated with the geomagnetic polarity time scale. Such deposits may be assigned to the Late Pliocene and Early Pleistocene (Matuyama reversed polarity chron: 2.6–0.78 Ma) or Late Pleistocene (Brunhes normal polarity chron: <0.78 Ma), or to one of the magnetic polarity subchrons within the Matuyama Chron.

The Late Cenozoic sedimentary record of the northern Canadian Cordillera is extensive and contains evidence of several glacial and interglacial periods. This evidence is based on studies of paleosols developed on multiple till sequences. The purpose of this study was to establish the suitability of the tills and associated paleosols for paleomagnetic analysis and to attempt a correlation of these deposits with the reference Quaternary magnetostratigraphic time scale. If indeed the sediments yield a stable magnetization and span a time period in excess of the last geomagnetic reversal (0.78 Ma), then we would expect to find both normally and reversely magnetized tills and paleosols. Furthermore, such a polarity record would be useful for regional correlation and would allow comparisons to be made with paleoclimatic records elsewhere.

In this study we present new observations of magnetic polarity of glacial deposits in the Mackenzie Mountains, which extend the correlations made on a lithological basis and provide a broad magnetostratigraphic framework for establishing the sequence of glaciations in the northern Canadian Cordillera.

Stratigraphy and sampling

Three sections in the Mackenzie Mountains (Fig. 1) provide the most complete stratigraphic record (Duk-Rodkin et al. 1996) and were selected for study (Katherine Creek, 64°57′N, 127°34′W; Little Bear River, 64°27′N, 126°45′W; Inlin Brook, 64°20′N, 126°37′W). They are located in a narrow belt of glacial deposits where till of both western (montane) and eastern (Laurentide) provenance occurs. At all three sections, a Pliocene pediment surface cut into bedrock or gravel is overlain by colluvium and multiple montane tills, each separated by paleosols. The sequences are capped by a Laurentide till (Fig. 2).

Samples were selected from sorted, fine-grained sediments, where present. Where not, collection was made from sorted coarser deposits that contained a fine-grained matrix, or from interstitial pockets within till or colluvium that was fine grained and appeared to be free of pebbles. In this manner, an attempt was made to sample all lithologic units at each of the sites. In particular, most tills revealed a well-developed paleosol at their surface, and where suitable both till and paleosol were sampled. Six to 10 specimens were collected from each sampling horizon (Fig. 2). This contrasts with traditional magnetostratigraphic studies where collections are usually made from outcrops or cores that are wholly suited for paleomagnetic analysis, and samples are taken in a quasi-continuous vertical sequence. Despite the sampling limitations in this study, polarities could be reliably assigned to most units.

Samples were collected by pressing plastic cylinders horizontally into the face. The cylinders (standard rock specimen dimensions) are made from shatterproof polycarbonate plastic. They have a sharp tapered lip that allows smooth penetration into the sediment face, and internal raised splines on the base and side that prevent movement of sediment inside the cylinder if shrinkage occurs and also serve as orientation marks.

At sampling horizons, a vertical face was cut into the unconsolidated deposits using a shovel and trowel. Sample azimuths were measured using a magnetic compass. Solar compass readings taken at each section showed good agreement. Altogether 216 samples were obtained, 131 from 10 units at Katherine Creek, 38 from 6 units at Little Bear River, and 47 from 4 units at Inlin Brook. The locations of the sampling horizons within each of the lithologic units, as well as the relative thickness of each unit, are shown in Fig. 2.

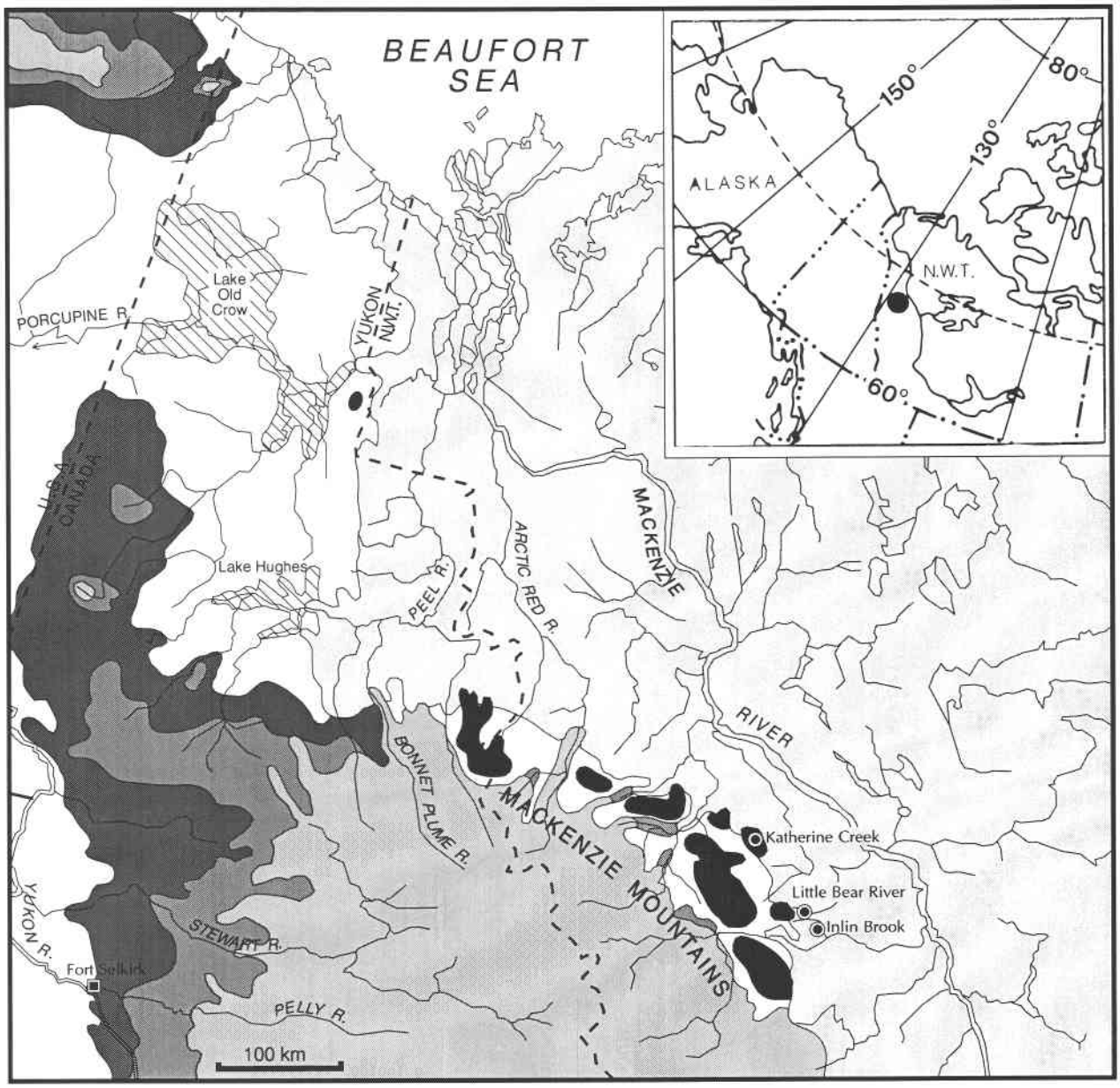
Paleomagnetism

Remanence measurements were made on an automated Schonstedt spinner magnetometer for samples with natural remanent magnetization (NRM) greater than about 2mA/m, and on a Geofyzika JR5A spinner magnetometer for the low-intensity samples. Stepwise alternating-field (AF) demagnetization was carried out using a Schonstedt GSD-5 with tumbler in peak fields up to 100 mT. Samples were demagnetized using 5–10 steps and directions determined by principal component analysis (PCA) (Kirschvink 1980).

Normal and reversed magnetizations occur at all three sections, and polarities are assignable to most colluvium, till, and paleosol units. Median destructive fields (MDF) range from 20 mT in unstable and generally uninterpretable horizons to >100 mT in horizons that reveal stable and coherent magnetization. Typically, the horizons with the best directional coherence have MDF's in the 60–100 mT range, indicating single-domain magnetite (probably detrital) as a dominant remanence carrier, but with a large component of hematite (probably as an alteration product). Tarnocai and Schweger (1991), Smith et al. (1986), and Duk-Rodkin et al. (1996) report that hematite, rather than the more highly oxidized minerals such as goethite, is found in the paleosols and contributes to their red hues.

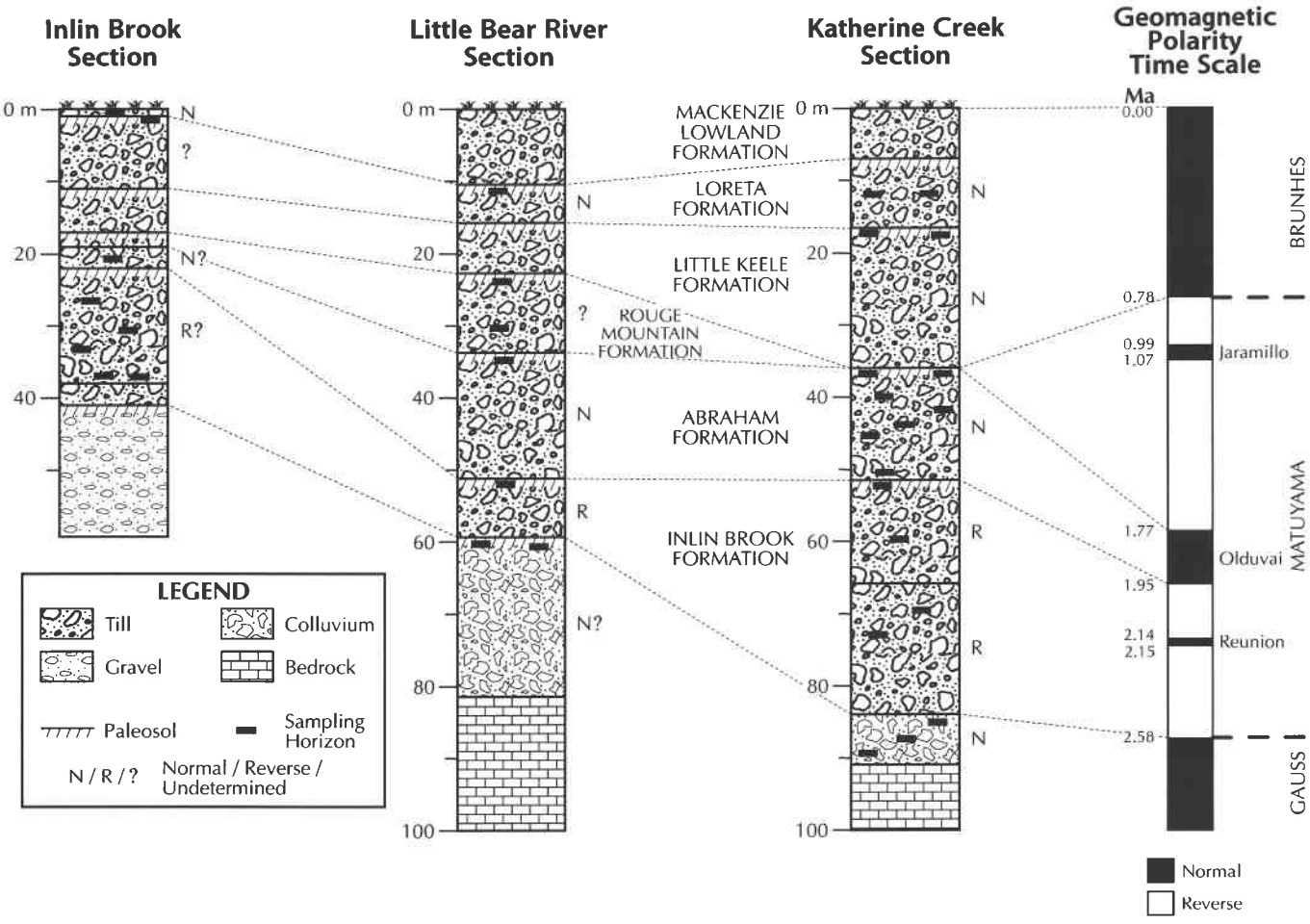
Since the demagnetization data seldom show evidence of multicomponent magnetization, the directions obtained by PCA have, in most cases, been obtained from lines that are forced through the origin of orthogonal plots. Figures 3*a* and 3*b* show examples of well-behaved normal and reverse polarity demagnetization curves, which are typical of a quarter of the sampled horizons. Another half of the collection produced noisy but unambiguous demagnetization trends toward the origin, similar to those illustrated in Figs. 3*c* and 3*d*. In the Little Bear River section, demagnetization trends that were not directed to the origin (e.g., Fig. 4) were frequently

Fig. 1. Location map showing Katherine Creek, Little Bear River, and Inlin Brook sections, and Laurentide and Cordilleran (montane) ice limits. Modified from Duk-Rodkin et al. (1996).



observed. These are indicative of multicomponent magnetization, and hence PCA linear segments were not forced through the origin. The remainder of the collection was noisy and uninterpretable. Polarity assignments were not made on the basis of single samples. Rather, we looked for coherence within sampling horizons. These data point to the limitations of paleomagnetic studies in highly variable and often poorly sorted or unsorted coarse-grained lithologies. Tills, for example, may contain pebbles that were not oriented in the Earth's ambient field at the time of deposition. The time-consuming task in such studies is to locate a sufficient number of fine-grained and pebble-free pockets to provide an

Fig. 2. Stratigraphic correlation of Katherine Creek, Little Bear River, and Inlin Brook sections. Correlations are based on polarity determinations and lithologic properties. The geomagnetic polarity reference time scale is based on the revised ages of Cande and Kent (1995). The quality of the paleomagnetic data on which polarity assignments are made is given in Tables 1 and 2.



adequate sampling of each lithologic unit.

In Table 1, NRM intensities (geometric mean) and Fisher-averaged directions of each unit are listed, and polarities defined. The mean directions are plotted in Fig. 5.

Katherine Creek section

The colluvium at the base of the Katherine Creek section (Fig. 2) rests directly on a Pliocene pediment cut into bedrock. Samples were collected from two fine-grained lenses within this unit and from a 0.5 m thick silty colluvial unit at the top. Most samples gave coherent normal directions after stepwise demagnetization, while a few were weakly magnetized and gave incoherent results.

The Inlin Brook Formation tills (Fig. 2) reveal generally coherent reversed directions, but a few samples have great scatter and were rejected. The paleosol had an order of magnitude weaker magnetization, and a much higher median destructive field (>100 mT), indicating a relatively high degree of alteration of magnetite to hematite. The directions were scattered, but clearly of reversed polarity.

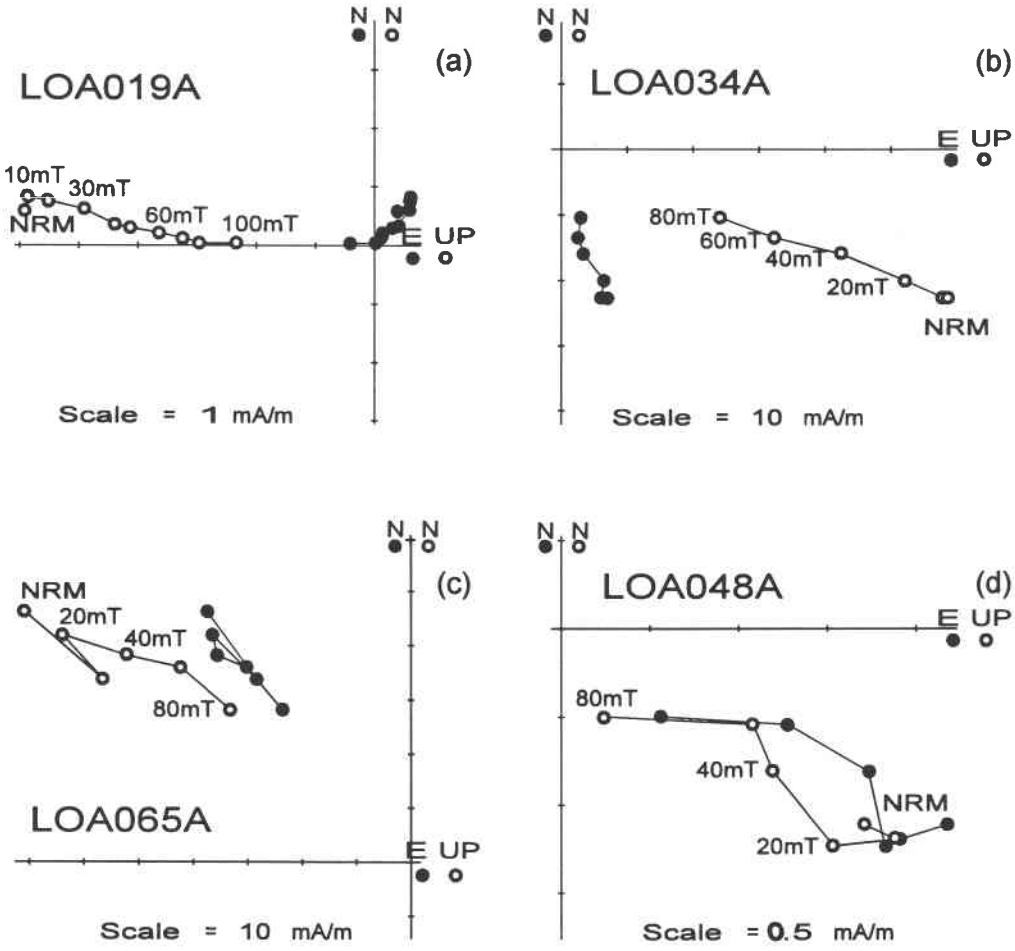
Four horizons were sampled within the till of the Abraham Formation, and all revealed good demagnetization characteristics and normal polarities (e.g., Fig. 3*a*). The paleosol developed on this till was sampled in two horizons, one showing scattered directions, the other coherent, but all of normal polarity. It is important to note that the paleosols of the Inlin Brook and Abraham formations record the same polarity as that of their parent material rather than the polarity of the overlying unit. Hence we can conclude that soil-forming processes, including weathering, leaching, and presumably the mobilization of irons, were arrested before the subsequent field reversal recorded in the overlying till. This has also been reported for till–paleosol sequences in the southern Canadian Rocky Mountains (Barendregt et al. 1991; Cioppa et al. 1995).

The till of the Little Keele Formation was too stony for paleomagnetic analysis; however, its paleosol was sampled at two sites, both yielding stable and coherent normal directions.

Two horizons in the Loreta Formation were sampled, each showing very high within-horizon coherence, but distinct normal directions, probably showing paleosecular variation. These distinct directions argue for the remanence being primary, since a complete remagnetization event would have affected the whole locality at a single time and produced identical directions.

The Loreta Formation is overlain by a till that contains granite clasts and thus provides evidence for the first Laurentide ice in the area. This till was too coarse for paleomagnetic

Fig. 3. Orthogonal plots illustrating typical demagnetization behaviour of the collection. The solid circles mark projections onto the horizontal plane, while open circles mark projections onto a vertical plane through the north. (*a*, *b*) Well-defined normal and reversed polarity behaviour. (*c*, *d*) Noisier but still acceptable normal and reverse polarity behaviour.



sampling, but at the Little Bear River section a chlorine date of ca. 22 ka on glacial erratics associated with till indicates it to be Late Wisconsinan (Duk-Rodkin et al. 1996).

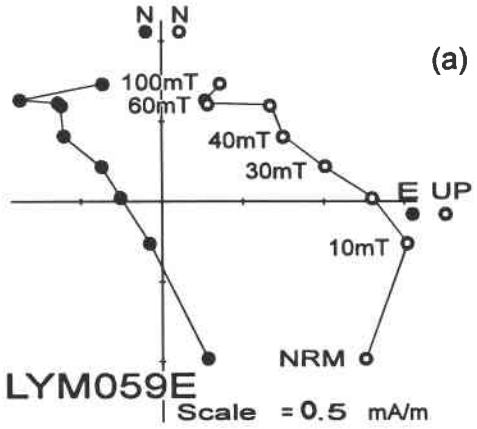
In summary, the Katherine Creek section exhibits a clear reversal stratigraphy that reveals a normal–reverse–normal sequence. Results are detailed in Table 2.

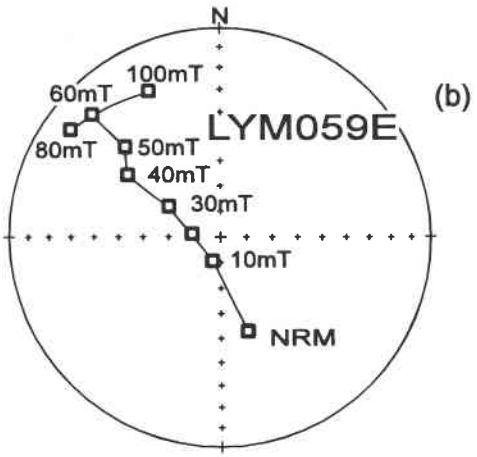
Little Bear River section

The Little Bear River section exposes a stratigraphic record similar to that found at Katherine Creek. Our purpose at this section was to confirm the results of the Katherine Creek section and extend the magnetostratigraphy over a broader geographic area. The Little Bear River section was clearly more oxidized (reflected in stronger colours and greater alteration overprints). Additionally this section contains much stonier tills and hence represented a greater challenge in sampling. Fisher averages for the coherent units are reported in Table 2. Polarity is defined for four units only, the others yielding incoherent directions.

The colluvium at the base of the Little Bear River section, like that at Katherine Creek, rests directly on a Pliocene pediment cut into bedrock. These sediments are weakly magnetized, and only two samples gave good normal directions, whereas the remainder gave noisy but generally normal directions. The paleosol of the Inlin Brook Formation exhibits a reversed polarity component of generally low stability, which is shown as upward-pointing lines not directed to the origin on orthogonal plots, and great-circle migration from the upper to the lower hemisphere on stereographic plots (e.g., Fig. 4). We interpret this behaviour to be due to a primary reversed component residing in detrital magnetite, which has been overprinted by oxidation alteration in the recent normal magnetic field. No high coercive force directions were isolated. The paleosol developed on the Abraham Formation shows good normal polarity and inclinations compatible with the sampling latitude. Till of the Rouge Mountain Formation shows generally noisy demagnetization curves, and some evidence of multicomponent behaviour, hence polarity is undetermined. A paleosol on the uppermost montane till (Loreta Formation) gave coherent normal magnetizations. The Little Bear River section was not as suitable for paleomagnetic characterization as the Katherine Creek section, but there is clear evidence of an older (reversed) Quaternary glaciation. Although the tills generally could not be sampled, the paleosol record correlates well with the Katherine Creek section. The results of this study suggest

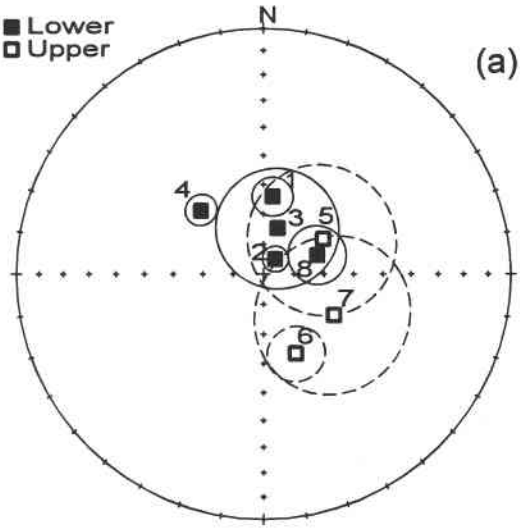
Fig. 4. Orthogonal plot (*a*), as in Fig. 3*b*, and corresponding stereograph (*b*) illustrating a characteristic reverse polarity component overprinted by a hard normal polarity component. Such behaviour is typical in the Little Bear River section, which has suffered recent oxidation, converting some detrital magnetite to hematite.

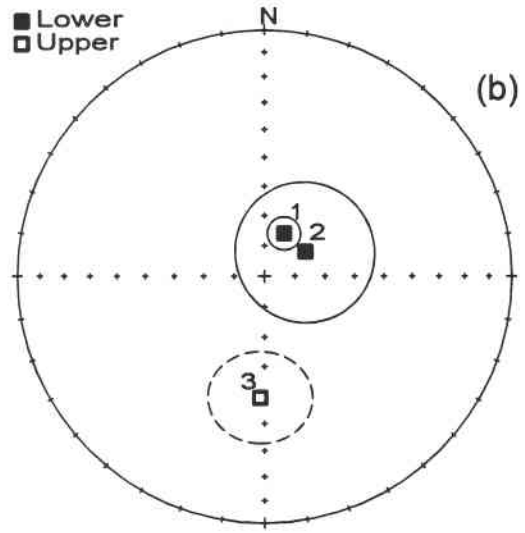




that paleosols reliably record the polarity of the underlying parent material.

Fig. 5. Stereographic plots of unit means for Katherine Creek section (*a*) and Little Bear River section (*b*). Numbers correspond to units listed in Tables 1 and 2. Circles are 95% ovals of confidence.





Inlin Brook section

Four tills were sampled from the Inlin Brook section (Fig. 2). Within the Inlin Brook Formation, the lowest, dark brown till could not be sampled because it was too gritty. A fine silty–sandy unit separating the dark brown till from the overlying red – chocolate brown till was sampled at two horizons, but gave generally noisy results. Two well-behaved samples, however, did show good reversed directions, suggesting that a correlation with the reversely magnetized lowermost till at Katherine Creek and Little Bear River sections is not unreasonable. The red – chocolate brown till was sampled at three horizons. The lowest horizon was in the till itself while the others were in successively higher sand pockets (0.50 m thick, 2.0 m wide) within the till. The till and first sand pocket were magnetically unstable. The second sand pocket yielded one normal and two reversely magnetized samples. The dark brown till and red – chocolate brown till are assigned to a single glaciation, there being no evidence of soil formation or an erosional hiatus at their contact. Although we cannot unequivocally assign a polarity to this formation, the bulk of the evidence would suggest that it is reversed. A sand pocket in the lower part of a till assigned to the Abraham Formation yielded normal magnetization but with anomalous westerly declinations. Again, a polarity assignment cannot be made with certainty, but the Abraham Formation is most likely normal. The uppermost of the montane tills (Loreta Formation) had very unstable magnetizations. Only the surface Laurentide till (Mackenzie Lowland Formation) had an unambiguous normal polarity record. The Inlin Brook section is not incompatible with the Katherine Creek and Little Bear River sections but adds little to the magnetostratigraphic constraints for this region of the Mackenzie Mountains.

Table 1. Katherine Creek section summary statistics.

Unit	N	NRM (mA/m)	D (°)	I (°)	k	α_{95} (°)	Polarity
1. Till of Loreta Formation	9	21.3	6.7	63.9	61.7	6.6	N
2. Paleosol on till of Little Keele Formation	14	5.0	37.9	83.6	84.8	4.3	N
3. Paleosol on till of Abraham Formation	12	11.7	17.0	74.0	5.6	20.2	N
4. Till of Abraham Formation	30	38.3	316.2	60.1	26.8	5.2	N
5. Paleosol on melt-out till of Inlin Brook Formation	5	1.7	59.8	−67.1	10.4	24.9	R
6. Melt-out till of Inlin Brook Formation	7	17.6	157.9	−61.1	41.8	9.4	R
7. Lodgement till of Inlin Brook Formation	10	5.7	121.0	−62.7	4.4	26.2	R
8. Colluvium over Pliocene pediment	13	2.3	70.2	71.1	19.1	9.7	N
Mean	100	7.7	337.2	73.3	9.1	4.9	

Notes: The lithologic units and formations are shown in Fig. 2, and unit mean directions are plotted in Fig. 5. N, number of samples; NRM, geometric mean of natural remanent magnetization intensity; D and I, declination and inclination of the mean remanence direction; k, Fisher precision parameter; α_{95}, 95% confidence interval; N, normal; R, reverse.

Table 2. Little Bear River section summary statistics.

Unit	N	NRM (mA/m)	D (°)	I (°)	k	α_{95} (°)	Polarity
1. Paleosol on till of Loreta Formation	3	9.8	24.6	74.5	498.0	5.5	N
Paleosol on till of Rouge Mountain Formation	7	3.6	—	—	—	—	Incoherent
Till of Rouge Mountain Formation	3	1.5	—	—	—	—	Incoherent
2. Paleosol on till of Abraham Formation	4	4.3	59.4	74.2	16.7	23.2	N
3. Paleosol on till of Inlin Brook Formation	8	2.0	182.0	−48.9	12.7	16.2	R
Paleosol on colluvium over Pliocene pediment	9	1.8	—	—	—	—	N?
Mean	15	3.3	12.9	62.6	10.9	12.1	

Notes: Mean of samples from coherent units only. Abbreviations as in Table 1.

Discussion

The paleomagnetic measurements outlined above clearly demonstrate that fine-grained deposits within glacial tills and associated paleosols do provide interpretable results on which magnetic polarity histories can be based. The normal–reverse–normal pattern established for these sections indicates that their sediments represent a very considerable span of time, and that units can be correlated over distances of hundreds of kilometres.

The reversely magnetized tills overlying the basal colluvium at both the Katherine Creek and Little Bear River sections provide evidence of a Late Pliocene glaciation in this region, a glaciation that predates the Brunhes–Matuyama reversal at 0.78 Ma. Based on lithologic properties and magnetization directions, the two reversely magnetized montane tills at the Katherine Creek section are thought to represent two stages of the same glaciation, and to be correlative with the lowermost till at the Little Bear River section. They are formally assigned to the Inlin Brook Formation (Duk-Rodkin et al. 1996).

The Inlin Brook Formation is overlain by several normally magnetized montane tills assigned to the Abraham, Little Keele, and Loreta formations. Based on the mature soil

development in these tills, as seen by their thick B horizons and significant clay accumulations, it can be assumed that each of the paleosols represents 30 ka of development (Duk-Rodkin 1996) and thus the formation boundaries represent a time span of at least this length. Several age assignments are possible for these normally magnetized tills and paleosols. Certainly the simplest option is to assign all to the Brunhes normal polarity chron, a period in the Quaternary that, based on the marine O^{18} record (Imbrie et al. 1984), is characterized by extensive cold periods.

There are, nevertheless, two other options that must be considered. They are the two normal subchrons within the Matuyama reversed polarity chron, namely the Jaramillo (1.07–0.99 Ma) and the Olduvai (1.95–1.77 Ma), both of which are considered to be of sufficient duration to contain a major glaciation. Arguments for considering the Jaramillo Subchron are found on the flank of Mount Edziza in northwestern British Columbia, where glacial diamicts interbedded with 1 Ma basalts show a reverse–normal–reverse sequence, of which the normal is unambiguously identified with the Jaramillo Subchron (Spooner 1994, Spooner et al. 1995). Mount Edziza is located in a physical setting and latitudinal belt similar to that of the Mackenzie Mountains and it is therefore possible that glacial sediments of Jaramillo age should be found at the Katherine Creek, Little Bear River, and Inlin Brook sections. Duk-Rodkin et al. (1996) argue that the normally magnetized Abraham Formation is best placed within the Olduvai subchron.

The Luvisol developed on the till of the Abraham Formation represents a long warm period, with mean annual temperatures higher than those of the present climate, and thus likely reflects climatic conditions more typical of the pre-Brunhes Chron. Also there is probably insufficient time within the Brunhes to accommodate five glaciations and interglaciations (Fig. 2). The Jaramillo Subchron is considered a less likely candidate because it is too short to include a glaciation as well as an interglacial period long enough to develop a Luvisolic soil.

Conclusions

The paleomagnetic record in the Mackenzie Mountains, together with the paleosol data provides strong evidence that this area was affected by multiple glaciations, spanning both the Matuyama and Brunhes chrons. These glaciations were separated by periods of soil formation, and based on soil properties, these periods were extensive, in the order of 30 ka. Despite these long periods, the primary magnetization of the paleosols was retained, and where measured, the polarity of each paleosol matches that of its till parent material. The large-scale changes in climate during the Late Pliocene and Quaternary, manifest as glacial–interglacial cycles in the terrestrial record of the Mackenzie Mountains, provide a unique opportunity to make a comparison with the marine record and to provide baseline data for paleoenvironmental reconstruction. Although the global O^{18} isotope record suggests that the Matuyama Chron was characterized by somewhat less severe, and certainly less extensive cold periods than the Brunhes Chron, it is clear that at least one glaciation did occur in the Mackenzie Mountains during the Late Pliocene.

Acknowledgments

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